

What makes paper displays a viable option is the fact that they're a lot cheaper to mass produce than the standard LCD, use several magnitudes less of battery power, and offer better screen visibility due to their higher contrast ratio—under direct sunlight, a paper display reads just like a newspaper or magazine, while an LCD is hardly visible.

Reduced battery consumption is perhaps the trump card that paper displays hold. Unlike conventional displays that use battery power continuously to display an image, paper displays use power only when they are drawing an image. So, if no change in the image is needed, the display needs no power at all, and one could even disconnect the power source altogether without the display going blank.

This means that even current batteries could last a lot longer on a device that uses a paper display. The decrease in manufacturing costs could yield even cheaper entry level phones, which do not offer advanced features such as video playback or cameras. Since a lot of the “no-frills” phones available today still come with monochrome displays that merely show caller ID and text messages, and therefore do not need high refresh rates, it seems logical that paper displays be used in such devices.

However, we're still waiting for the day that paper displays capable of displaying video are developed. That day might spell doom for the now-ageing LCD technology.

